WEEK 5 · DAY 1 · INSTRUCTOR ONLY

Quiz 3 Answer Key

Machine learning · shuffled version

Instructor only. Built from the original `Time-series/MCQ/week-05-mcq.csv` (that file is unchanged), with 8 questions edited: see *Changes from the original* below. Options were reshuffled with a fixed seed (5103) so the correct letters are spread out: multiple-choice answers sit at A ×7, B ×6, C ×6, D ×7 (in the original, most answers were option B). The 4 True/False questions keep their original order.

#	Answer	Correct option	What it tests	Level
1	**A**	It can notice when the relationship between channels breaks	What ML buys you	beginner
2	**D**	fit, predict (or transform), score	The one pattern	beginner
3	**D**	Rows are hours, columns are channels	The X matrix	beginner
4	**A**	Without scaling, channels with large ranges dominate distance-based methods	Why scale	intermediate
5	**B**	The scaler fitted on training rows is the one applied at prediction time	Pipelines	intermediate
6	**B**	False	Unsupervised	beginner
7	**C**	Something in the code is wrong and needs checking	Verifying generated code	intermediate
8	**C**	Four	Faults one channel can see	beginner
9	**C**	It needs few random cuts to isolate it	Isolation intuition	intermediate
10	**A**	A correlation break	EV02	intermediate
11	**D**	More anomalous	Decision function	beginner
12	**A**	About 44	Contamination	intermediate
13	**A**	Finding the operating modes such as sunlit and eclipse	Clustering use	intermediate
14	**B**	Per-channel z-scores against the training baseline	Explaining a flag	intermediate
15	**A**	True	When the baseline wins	advanced

#	Answer	Correct option	What it tests	Level
16	**B**	Each row is judged on its own unless features carry history such as lags and rolling stats	Why engineer features	intermediate
17	**D**	Amplifier temperature relative to its RF drive level	temp_per_drive	beginner
18	**D**	The model also flags every hour from October to December, so it is useless in service	EV04 in sigma	advanced
19	**C**	df[c].shift(-1)	Features that cheat	intermediate
20	**C**	Could I compute this at 14:00 using only what I knew at 14:00?	The feature test	beginner
21	**B**	Duplicates double-weight that channel because split dimensions are chosen at random	Redundant features	advanced
22	**B**	The longest window is 24 hours	Rows lost	beginner
23	**C**	Test hours have near-identical neighbours in the training set	Why random lies	beginner
24	**A**	assert tr.index.max() < te.index.min()	The assertion	intermediate
25	**D**	At least as long as the longest feature window	The embargo	intermediate
26	**D**	It gives a spread of scores instead of one possibly lucky number	Rolling origin	intermediate
27	**A**	True	Training only grows forwards	beginner
28	**B**	Same held-out period and the same scorer for both detectors	Fair comparison	beginner
29	**A**	From a period operations independently believe was quiet, stated in the write-up	Anomaly-specific trap	advanced
30	**A**	True	Recommending against your model	intermediate

Changes from the original

The same fixes are in the weekly question banks (`week-N/MCQ/`).

- **Q1** (reworded): Week 4 of the original course is Week 3 here.
- **Q4** (corrected): The option "It does not matter for isolation forest" was also true (the forest's scores with and without scaling correlate 0.99999999), so the question had two right answers.
- **Q6** (reworded): Week 5 of the original course is Week 4 here.
- **Q8** (corrected): The original answer "Two" follows the label file, which only checks array current. A 3-sigma chart on each raw channel, on steady-sun hours, catches four; only EV04 needs two channels.
- **Q11** (reworded): It said "in the code book", which participants never see.

- **Q15** (reworded): Week 4 of the original course is Week 3 here.
- **Q18** (corrected): The original answer "About 22 sigma" does not reproduce: the recipe gives 28.9 sigma, and the same feature flags all 2,208 hours of October to December. The question now asks the lesson the course teaches.
- **Q28** (reworded): It named the original course's Weeks 4 and 5.

Marking it together (10 minutes)

1. Read out the letters; everyone marks their own paper.
2. Ask for hands up on each question. Any question that more than a third of the room got wrong, explain it again with the matching day's slide.
3. Map each topic back to its day: the scikit-learn workflow → W4D2 · isolation forest and clustering → W4D2 · feature engineering from time series → W4D3 · temporal train-test discipline → W5D1.

Quick topic tally

Topic	Questions
the scikit-learn workflow	1, 2, 3, 4, 5, 6, 7
isolation forest and clustering	8, 9, 10, 11, 12, 13, 14, 15
feature engineering from time series	16, 17, 18, 19, 20, 21, 22
temporal train-test discipline	23, 24, 25, 26, 27, 28, 29, 30